Computational Statistics and Simulation

Worksheet 1: Generation of Pseudo-Random Numbers

Academic year 2025/26

1. Generate values from a Normal distribution with mean 2 and variance 4.
   (a) Compute the sample characteristics: mean, median, standard deviation and quantiles.
   (b) Evaluate the probability density function and the cumulative distribution function at the generated points and plot them.
   (c) Draw the histogram of the generated sample.
   (d) Consider a sequence of values appropriate to the generated sample.
      i. Evaluate the p.d.f. of the $N(2, 4)$ distribution on that sequence and plot it.
      ii. In a single plot, show the histogram of samples generated with $n = 100$ together with the p.d.f. evaluated on the sequence you created; redo the plot using a sample of size 1000.

2. Generate values from a Binomial $(100, 0.25)$ distribution.
   (a) Compute sample characteristics of the generated values and evaluate the p.m.f. and the c.d.f. at several points.
   (b) Draw the bar plot of 5000 generated values using $B(10, 0.25)$, $B(50, 0.25)$ and $B(100, 0.25)$, and add to the same plot the p.m.f. values evaluated on a sequence of points (using the binomials mentioned above).
   (c) Consider the $B(100, 0.25)$ distribution. Generate 5000 values from an $N(25, \text{var} = 18.75)$ distribution and compare the two distributions graphically.

3. A random variable X is said to have a type I Pareto distribution with parameters α and β ($\alpha > 0$, $\beta > 0$) — written $X \sim Pa(\alpha, \beta)$ — if its probability density function has the form

$$f(x \mid \alpha, \beta) = \alpha\beta^{\alpha}x^{-(\alpha+1)}, \qquad x \geq \beta,$$

with $E(X) = \dfrac{\alpha\beta}{\alpha - 1}$ and $\text{Var}(X) = \dfrac{\alpha\beta^2}{(\alpha-1)^2(\alpha-2)}$; assume $\alpha = 3$, $\beta = 1.5$.

   (a) Without using the R functions specifically implemented for the Pareto distribution, write a set of R functions that returns, for this Pareto distribution:
   - the value of the probability density function (p.d.f.) at a given x (x may be a vector of values at which the p.d.f. is to be evaluated);
   - the value of the cumulative distribution function (c.d.f.) at a given x (x may be a vector of values at which the c.d.f. is to be evaluated);
   - the quantiles of probability p (p may be a vector of probabilities);
   - n pseudo-random numbers from that distribution.

   Hint: solve the previous exercise using the R commands for the Pareto distribution (`pareto`), installing the `extremefit` package.
   (b) Varying the parameters α and β, plot the probability density function of a $Pa(\alpha, \beta)$. Comment.
   (c) It is known that this distribution has a mean if $\alpha > 1$ and a variance if $\alpha > 2$. Compute the sample means and variances of simulated samples in the following cases: when $\alpha \leq 1$ and/or $\alpha \leq 2$. Comment on the results.

4. A random variable X is said to have a logistic distribution with parameters α, β ($\alpha \in \mathbb{R}$, $\beta > 0$) — written $X \sim Lo(x \mid \alpha, \beta)$ — if its probability density function is given by

$$f(x \mid \alpha, \beta) = \frac{\beta^{-1} \exp\left\{-\left(\frac{x-\alpha}{\beta}\right)\right\}}{\left[1 + \exp\left\{-\left(\frac{x-\alpha}{\beta}\right)\right\}\right]^2}, \qquad x \in \mathbb{R}.$$

Without using the R commands for the logistic distribution and taking $\alpha = 1$, $\beta = 2$, write a set of R functions that returns:

(a) the value of the probability density function (p.d.f.) at a given x (x may be a vector of values at which the p.d.f. is to be evaluated);

(b) the value of the cumulative distribution function (c.d.f.) at a given x (x may be a vector of values at which the c.d.f. is to be evaluated);

(c) the quantiles of probability p (p may be a vector of probabilities);

(d) n pseudo-random numbers from that distribution.

Hint: solve the previous exercise using the R commands for the logistic distribution (`logis`).

5. For X, consider the Beta model with parameters $(2, 2)$, whose probability density function is $f(x) = 6x(1-x)$, $0 < x < 1$. The aim is to generate values of this variable using the Rejection Method, choosing the uniform $(0, 1)$ model as the proposal function.

(a) On average, how many $U(0, 1)$ pseudo-random numbers are needed to generate 1000 values of that random variable?

(b) Generate 1000 values, including a counter for the number of iterations actually needed to obtain the 1000 observations, and compare it with the expected number of iterations.

(c) Compare pointwise the empirical percentiles, sample mean and sample variance with the corresponding theoretical values of the $Be(2, 2)$ distribution.

(d) Plot the histogram of the n generated values and superimpose the p.d.f. curve of the $Be(2, 2)$.

(e) Increase the number of generated values to $n = 10000$ and comment on the results obtained previously.

6. Consider the following mixtures of Gamma distributions:

(a) $0.3\, Ga(1.5, 3) + 0.7\, Ga(5, 3)$;

(b) $\sum_{i=1}^{5} \theta_i f_{Ga(3,\theta_i)}$, where $\theta_1 = 0.1$, $\theta_2 = 0.2$, $\theta_3 = 0.2$, $\theta_4 = 0.3$, $\theta_5 = 0.2$.

Generate values from each mixture and plot them.

7. A discrete random variable is said to have a Beta-Binomial distribution with parameters α, β and n, written $X \sim Betabin(n, \alpha, \beta)$, if its probability mass function has the form

$$f(x \mid \alpha, \beta, n) = c \binom{n}{x} \Gamma(\alpha + x)\Gamma(\beta + n - x), \quad x = 0, \ldots, n; \qquad c = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)\Gamma(\alpha + \beta + n)}.$$

(a) Show that this distribution is generated by mixing a binomial distribution with a beta distribution, that is,

$$f(x \mid \alpha, \beta, n) = \int_0^1 \binom{n}{x} y^x (1-y)^{n-x} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} y^{\alpha-1}(1-y)^{\beta-1}\, dy, \quad x = 0, \ldots, n.$$

(b) Taking the parameter vector $(\alpha, \beta, n) = (10, 22, 4)$, generate values from this distribution.

8. Generate samples (x_i, y_i) from a bivariate distribution with joint p.d.f.

$$f_{(X,Y)}(x, y) = \begin{cases} 2, & 0 < x < y < 1, \\ 0, & \text{other } (x, y) \in \mathbb{R}^2, \end{cases}$$

and plot the generated observations (considering different sample sizes).